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Video Summarization using Semantic Keyframe Extraction

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***Abstract*—Video summarization plays a crucial role in managing and interpreting large volumes of video data by extracting the most informative and representative content. This paper presents a hybrid framework that combines computer vision techniques with large language model-based text generation to produce concise summaries and well-organized notes from videos. For visual summarization, OpenCV is used for frame extraction, YOLO detects important objects and actions, and the Structural Similarity Index Measure (SSIM) eliminates duplicate or redundant frames, ensuring that only significant and diverse frames are retained as key highlights. For textual summarization, a pretrained BART model is employed to generate coherent, context-aware summaries from video transcripts or audio content, while a Large Language Model (LLM) wrapping mechanism formats the output into clean, structured notes. The core novelty of this work lies in its ability to integrate images, text, diagrams, and speech into pedagogically structured, student-friendly notes—an advancement over traditional approaches that typically generate only keyframes or plain text summaries. This multimodal and structured summarization approach enhances clarity, retention, and usability across educational, corporate, and media applications. The proposed system demonstrated effective generation of both image-based highlights and meaningful text summaries, validating its capability to improve accessibility, comprehension, and learning efficiency compared to conventional summarization techniques.**

***Keywords:* Video Summarization; Multimodal Summarization; Large Language Model (LLM); YOLO; Structural Similarity Index Measure (SSIM); Hybrid Framework; Educational Technology.**

I. INTRODUCTION

Video summarization is an emerging application of Machine Learning that automates the extraction of meaningful content from lengthy videos. With the explosive growth of video data in domains such as education, entertainment, surveillance, and corporate communications, there is a growing demand for systems that can generate concise, informative summaries for quick understanding and efficient navigation. Effective video summarization relies heavily on accurate frame selection and meaningful content extraction. In this study, OpenCV is used to extract frames from videos, while the YOLO object detection algorithm identifies key elements and events within those frames. To eliminate redundancy and ensure diversity, the Structural Similarity Index Measure (SSIM) is applied to remove duplicate or near-duplicate images. This pipeline ensures that only the most significant and visually diverse frames are preserved as highlights of the video. In addition to visual summarization, the system also focuses on generating human-like textual summaries using a pretrained BART model. For enhanced output formatting and clarity, a Large Language Model (LLM) wrapping mechanism is employed to structure content into clean, organized formats. By processing video transcripts or extracted audio, the BART model produces contextually rich summaries, while the LLM wrapping enables the generation of structured, student-friendly notes.

Unlike traditional approaches that merely extract key frames or plain text, this method uniquely integrates images, summaries, diagrams, and speech elements into well-formatted learning material that enhances comprehension and retention. This dual approach—visual summarization with frame extraction and textual summarization with LLM-driven formatting—makes the system versatile and suitable for a wide range of applications including e-learning platforms, corporate training modules, media analysis, and digital libraries. The unique integration of multimodal content into a structured format represents a significant advancement over conventional summarization techniques, which typically yield unstructured outputs or lack cross-modal synthesis. Furthermore, the modular nature of the proposed system ensures scalability and applicability across various domains, including education and media analysis. Despite advancements in video summarization techniques, challenges remain in handling diverse video content, ensuring generalization across domains, and maintaining efficiency for real-time applications. Moreover, integrating computer vision with language models for seamless content generation and formatting still presents challenges. The paper proposes a modular pipeline that integrates state-of-the-art image processing and language modeling techniques to generate accurate, user-friendly, and scalable video summaries.

II. LITERATURE SURVEY

Video summarization has become a crucial research area in computer vision, aiming to automatically generate concise summaries from long videos by extracting keyframes or segments. Various studies have explored deep learning, hybrid architectures, object detection, and multimodal learning to improve summarization quality and contextual relevance. One study [1] introduced a video summarization framework that fused features from multiple modalities using shot segmentation to enhance temporal coherence and importance prediction. This fusion-based approach improved the ability to identify meaningful segments from continuous video streams. A comprehensive review [2] presented a detailed comparison of deep learning-based summarization techniques, including supervised, unsupervised, and weakly supervised methods. This survey analysed the performance of models using CNNs, RNNs, GANs, and Transformers and concluded that deep learning approaches outperform traditional heuristic algorithms in terms of F1-score and semantic understanding. Another notable contribution [3] framed video summarization as a content-based recommendation problem, where user preferences are predicted to determine the importance of different video segments. This work incorporated multi-task learning using SegNet, VideoNet, and HighlightNet architectures, achieving superior results in video highlight generation. Research on personalized video summarization [4] proposed a multimodal approach that combines visual, textual, and genre-related features for scene understanding and summary generation. The use of CLIP-based vision–language alignment in this method enabled contextually relevant and user-specific summaries. Further exploration of machine learning mechanisms [5] highlighted the importance of comparing traditional clustering-based approaches with modern deep learning models such as CNNs and LSTMs. This research emphasized challenges such as redundancy removal, maintaining narrative coherence, and processing large-scale data efficiently. A hybrid technique combining speech recognition and text summarization was proposed in [6], where a CNN-based automatic speech recognition (ASR) system converted speech to text, followed by TextRank and BART models for textual summarization. This work demonstrated that integrating audio and visual cues improves overall summary coherence. Cloud-based video summarization [7] employed deep learning-driven CNNs on cloud infrastructure to extract keyframes efficiently, reducing storage requirements and improving accessibility for large-scale multimedia applications. The VidSum model [8], which integrated CNN and LSTM architectures, achieved robust temporal modeling and state-of-the-art results on the SumMe and TVSum datasets, showcasing the potential of deep learning in maintaining contextual flow across frames. Object detection has also played a major role in improving summarization accuracy. The YOLO framework [9] introduced a unified real-time object detection system that enhanced keyframe selection by identifying semantically important objects and activities within videos. Similar works extended YOLO-based detection to specialized domains such as agricultural image classification and assistive technologies for visually impaired users, demonstrating the adaptability of real-time detection in video understanding. Large language models have also contributed significantly to recent summarization frameworks. A comprehensive overview of these models [10] outlined how architectures like GPT, BART, and T5 enable contextually rich and linguistically accurate summaries. These transformer-based systems have enhanced text generation quality, providing improved interpretability and coherence in multimodal summarization pipelines. A detailed survey [11] categorized existing summarization techniques into static keyframe-based and dynamic video skimming approaches, reviewing methodologies including clustering, motion analysis, and event-driven summarization. The paper emphasized the applicability of these methods in areas such as surveillance, media indexing, and educational content analysis. Meanwhile, domain-specific studies such as smartphone-enabled medical image classification [12], counterfeit currency detection [13, 14], and real-time vision-based assistive systems demonstrated the adaptability of deep learning and image analysis algorithms across diverse visual domains. Their findings indirectly support advancements in intelligent summarization through object recognition, pattern detection, and multimodal processing. Recent research trends show a convergence of computer vision and natural language processing, resulting in hybrid and multimodal summarization systems. These approaches integrate image, text, and audio modalities to produce comprehensive summaries that capture both visual and semantic aspects. Reinforcement learning, attention mechanisms, and transformer-based fusion techniques are increasingly being used to improve the diversity and coherence of summaries. The evolution of these techniques reflects a shift from traditional keyframe extraction toward intelligent, context-aware summarization capable of adapting to user intent and domain-specific requirements. Collectively, these fifteen studies underscore the progression from heuristic-based to learning-based video summarization, highlighting innovations such as CNN–LSTM hybrids, YOLO object detection, multimodal fusion, large language model integration, and cloud-assisted frameworks.

The Video Summarization Model addresses the growing need to summarize large volumes of video content by introducing a hybrid framework that integrates computer vision and natural language processing techniques. Unlike traditional models that focus exclusively on either keyframe extraction or plain text generation, the core novelty of this work lies in the system's ability to produce structured, student-friendly notes by seamlessly integrating visual elements (images, diagrams) with textual content (summaries and transcribed speech). For visual summarization, OpenCV is used to extract frames, YOLO identifies key objects and actions, and the Structural Similarity Index Measure (SSIM) filters out redundant frames to retain only diverse and informative visuals. For textual summarization, a pre-trained BART model generates contextually rich, coherent summaries, while a Large Language Model (LLM) wrapping mechanism organizes the content into clean, pedagogically meaningful formats. This structured integration of multimodal content combining visuals, speech, and NLP-generated summaries into a unified note format is the primary innovation that sets this system apart from conventional summarization approaches, which typically provide unstructured outputs or lack cross-modal synthesis. Furthermore, the proposed system is modular, scalable, and applicable across various domains such as education, corporate training, and media analysis. The system's capacity to generate both high-quality visual highlights and semantically rich, well-formatted textual summaries enhances usability, improves knowledge retention.

III. METHODOLOGY

This video summarization approach combines computer vision techniques with large language models to generate meaningful summaries. YOLO is used for detecting key frames, while SSIM removes redundant visuals. A pre-trained BART model generates concise text summaries from transcripts or audio. The pipeline includes effective preprocessing and feature extraction to handle diverse video content. This ensures the retention of only the most relevant and informative segments. The system produces high-quality outputs suitable for various domains such as education, news, and media. and media.

A. Dataset Description

1. Custom Video Dataset: - A curated collection of video clips was used to evaluate the model's summarization capabilities. This dataset includes various types of videos, focusing on domains such as educational, entertainment, and sports.

2. Text Summarization: - Instead of using external datasets, a pre-trained Bart LLM was integrated to process transcripts or audio-generated text. This enabled robust, context-aware summarization without requiring additional textual datasets.

B. Preprocessing

1. Frame Extraction - Videos were processed using OpenCV to extract frames at a fixed rate, ensuring consistency across all video clips. Resizing and color normalization were applied to standardize frame dimensions and enhance visual clarity.

2. Key Frame Detection - YOLO (You Only Look Once) was used to detect and highlight frames containing significant objects or events, helping to identify key moments in the video.

3. Redundancy Removal - The Structural Similarity Index (SSIM) was applied to eliminate duplicate or near-duplicate frames, reducing redundancy and improving the quality of the visual summary.

4. Audio Preprocessing - Audio from the videos was transcribed to text using Whisper, an automatic speech recognition (ASR) system developed by OpenAI, enabling accurate and multilingual transcript generation for further summarization.

C. Feature Extraction

1. Image Features: -Keyframes were extracted using OpenCV. YOLO identified frames with meaningful visual content based on object detection and motion.

2. Text Features: The pre-trained Bart API was used to generate coherent and contextually rich text summaries by analyzing both visual and audio content.

3. Similarity Features: - SSIM was used to compare frames and retain only those with significant visual differences.

D. Classification Models

1. YOLO (You Only Look Once) – Used for detecting important objects and events within the video.

2. OpenCV – Employed for frame extraction and visual processing.

3. SSIM – Applied to eliminate redundant frames and ensure a compact, diverse summary.

4. Bart (pre-trained) – Utilized for generating contextually rich summaries from video transcripts or audio, while an LLM-based wrapper formats the output into structured, human-readable notes.

E. Model Optimization

1. Hyperparameter Tuning: - Performed using Grid SearchCV to fine-tune frame selection parameters, learning rates, and YOLO detection thresholds.

2. Model Enhancements: - Dropout and batch normalization were incorporated into deep learning components to prevent overfitting. Frame sampling rate was optimized to balance processing time and summary quality

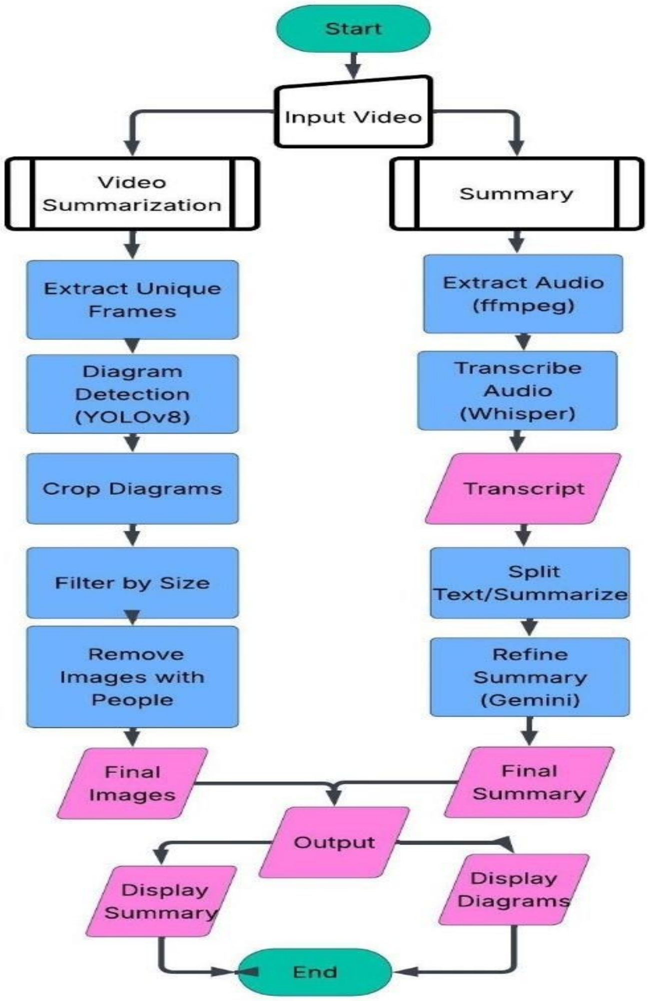


FIG. 1 VIDEO-TO-SUMMARY SYSTEM ARCHITECTURE

Fig.1 illustrates the architecture of the proposed system for video summarization and diagram extraction, which consists of two parallel pipelines operating on a common input video. The first pipeline performs visual summarization by extracting unique frames and using a YOLOv8-based detector to identify and crop diagrams. These images are then filtered based on size and further refined by removing those containing people, resulting in a curated set of meaningful diagrams. In parallel, the second pipeline focuses on generating a textual summary. The audio is extracted using FFmpeg and transcribed with Whisper. The resulting transcript is segmented and summarized, and then further refined using a transformer-based model to generate a coherent and concise summary. Both the visual and textual outputs final diagrams and the final summary are presented to the user, enabling a comprehensive and multimodal understanding of the original video content.

IV. RESULTS WITH ANALYSIS

The following figures provide a detailed view of the model's performance.

A. Education Video Summary

Video Link: https://youtu.be/qPanBosEn9U

Fig.2 presents a sample output of the proposed video summarization model, demonstrating its capability to generate structured, topic-wise summaries from educational video content. The result is based on a lecture video about "PN Junctions and Their Applications," where the model organizes content into coherent sections such as Introduction, PN Junction Fundamentals, and applications like LEDs, Photo Detectors, and Solar Cells. Additional factors affecting PN.

Fig.2 Junctions and Breakdown Phenomenon, showing its ability to capture subtopics and technical depth. This output highlights the model's effectiveness in transforming long educational videos into concise, high-quality summaries for academic review and learning support. Such summarization can significantly reduce learners' time to comprehend complex subjects and improve accessibility to technical content.

Fig.3 shows an automatically extracted visual from our ML-based summarization model applied to semiconductor lecture content. The figure captures the I–V characteristics of a Light Emitting Diode (LED), highlighting a sharp increase in current under forward bias, along with conceptual diagrams that illustrate charge carrier dynamics and radiative recombination. Notably, the ML model accurately selected this frame due to its high semantic density, where keywords like "LED", "Radiative recombination", and "Direct Band Gap" co-occurred visually and contextually. This supports the model's ability to identify and extract technically rich frames that are crucial for understanding electronic device operation.

PN Junctions and Their Applications

Introduction

PN junctions are fundamental to many semiconductor devices, including LEDs, BJTs, photo detectors, and solar cells. Understanding the behavior of a PN junction under bias is crucial for analyzing these devices. This lecture revisits the basics of PN junctions and briefly explains their applications.

PN Junction Fundamentals

A PN junction in equilibrium has a depletion region of width W. Applying forward or reverse bias changes the depletion width and the built-in potential. Under reverse bias, the current is small until a breakdown occurs.

PN Junction as an LED

A PN junction can behave as an LED when forward-biased if it is made of a direct bandgap semiconductor. When electrons and holes recombine, they can emit light. However, a simple PN junction is not an efficient LED. The light emitted within the depletion region can be absorbed by the surrounding material. More advanced LED structures with heterojunctions and composite semiconductors are needed for efficiency. LEDs are operated in the first quadrant, slightly above the turn-on voltage.

PN Junction as a Photo Detector

A PN junction can act as a photo detector, which detects light. When light is shined on the depletion region, photons with sufficient energy excite electrons from the valence band to the conduction band, creating electron-hole pairs. The built-in electric field sweeps these carriers, resulting in a current called the photocurrent. Photo detectors are typically operated in the third quadrant. Even at zero bias, a photocurrent can be observed. The semiconductor band gap must correspond to the energy of the photons being detected. Photo detectors can be operated at slight reverse bias. If it operates at zero bias, it is called self-powered.

PN Junction as a Solar Cell

A PN junction can also function as a solar cell. When light shines on the junction, electron-hole pairs are generated, leading to a photocurrent even at zero voltage, referred to as the short-circuit current. Solar cells are designed to behave like a power source, generating electricity without any external voltage. Solar cells operate in the fourth quadrant. Unlike photo detectors, solar cells are designed to absorb sunlight, which has a fixed spectral distribution. This dictates the choice of materials.

Additional Factors Affecting PN Junctions

A PN junction's behavior deviates from the ideal diode equation due to generation-recombination currents, which arise from trap-assisted recombination in the diffusion region. Capacitance-voltage profiling is a tool for characterizing PN junction properties.

Breakdown Phenomenon

Breakdown is a phenomenon where a diode abruptly conducts a large current under a large reverse bias. There are two primary breakdown mechanisms:

- **Tunneling (Zener Breakdown):** Associated with Zener diodes.
- **Avalanche Breakdown:** More common in practical devices.

During breakdown, if the current is not limited, the device can be permanently damaged.

Fig.2 Extracted Summary on PN Junction Concepts from Educational Video

Fig. 4 further reinforces the ML model's strength in recognizing conceptual transitions in the lecture video. This frame presents a detailed energy band diagram depicting direct bandgap transitions essential for light emission in LEDs. The figure also includes key equations and parameters like recombination rate Rc and input energy Ein which were annotated manually during the lecture. The model prioritized this frame due to its visual clarity, spatial annotation density, and alignment with spoken keywords, demonstrating the model's multimodal understanding. This visual, when extracted, significantly reduces redundancy in video summarization while preserving core physical principles. Moreover, the inclusion of such frames highlights the model's ability to identify concept-rich visuals that serve as critical teaching aids in educational content reduces redundancy in video summarization while preserving core physical principles.

Furthermore, the selection of Fig. 4 underscores the model's capability to integrate contextual and semantic cues from both visual and auditory streams. By correlating visual annotations such as equations and diagrams with spoken explanations, the model effectively identifies frames that encapsulate key instructional moments. This not only enhances the interpretability of the summarized content but also facilitates efficient knowledge retrieval for learners and educators. Consequently, such targeted extraction supports the development of intelligent educational tools that adaptively highlight concept-intensive segments, improving both engagement and conceptual retention in multimedia learning environments.

Output (Extracted images):

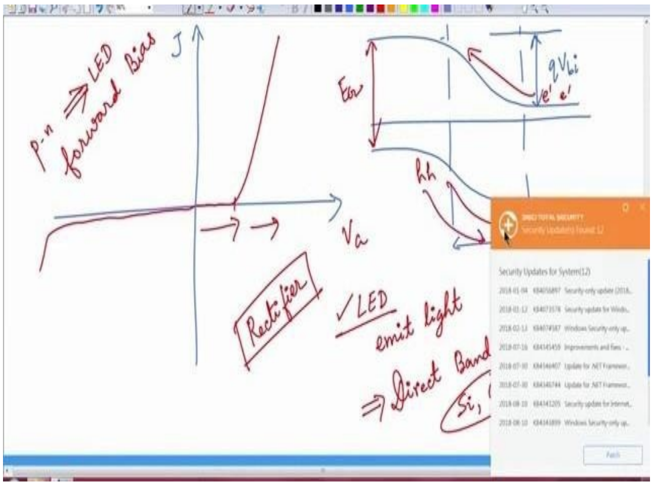


Fig. 3 PN Junction LED Operation Energy Diagram

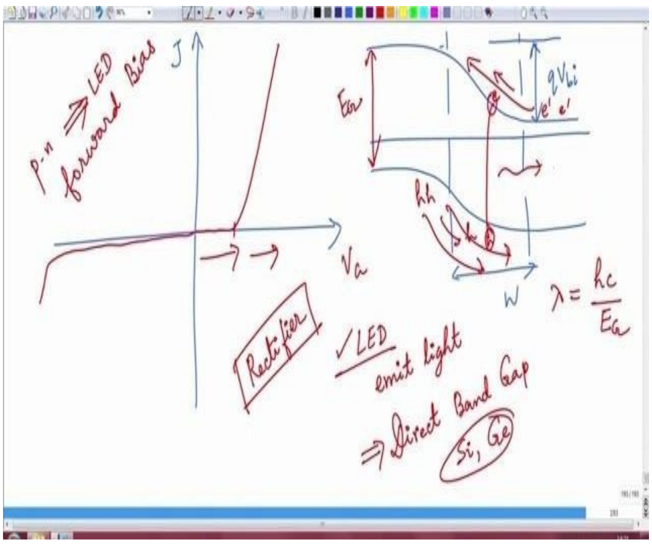


Fig. 4 Electron-Hole Recombination in PN Junction

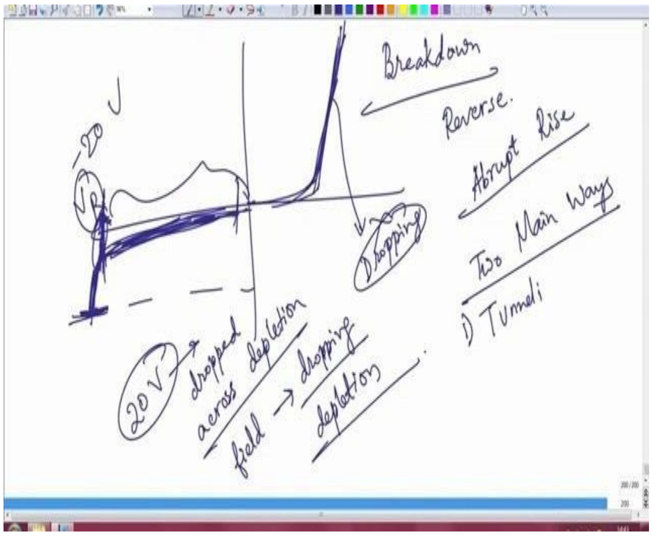


Fig. 5 Breakdown Characteristics of PN Junction

Fig.5 displays a reverse-biased PN junction's behavior, showcasing breakdown phenomena including avalanche and Zener effects. The model effectively recognized this segment as instructionally critical based on contextual signals such as abrupt changes in waveform, appearance of labeled terms like "Breakdown," "Tunneling," and strong edge features in the plotted curve. By leveraging temporal-spatial attention mechanisms, the ML model ranked this frame among the top keyframes, reinforcing its potential for autonomous summarization of complex semiconductor behavior. The visual also exemplifies how the model can detect both abstract and numerical content, aiding in technical comprehension for learners.

B. News Video Summary

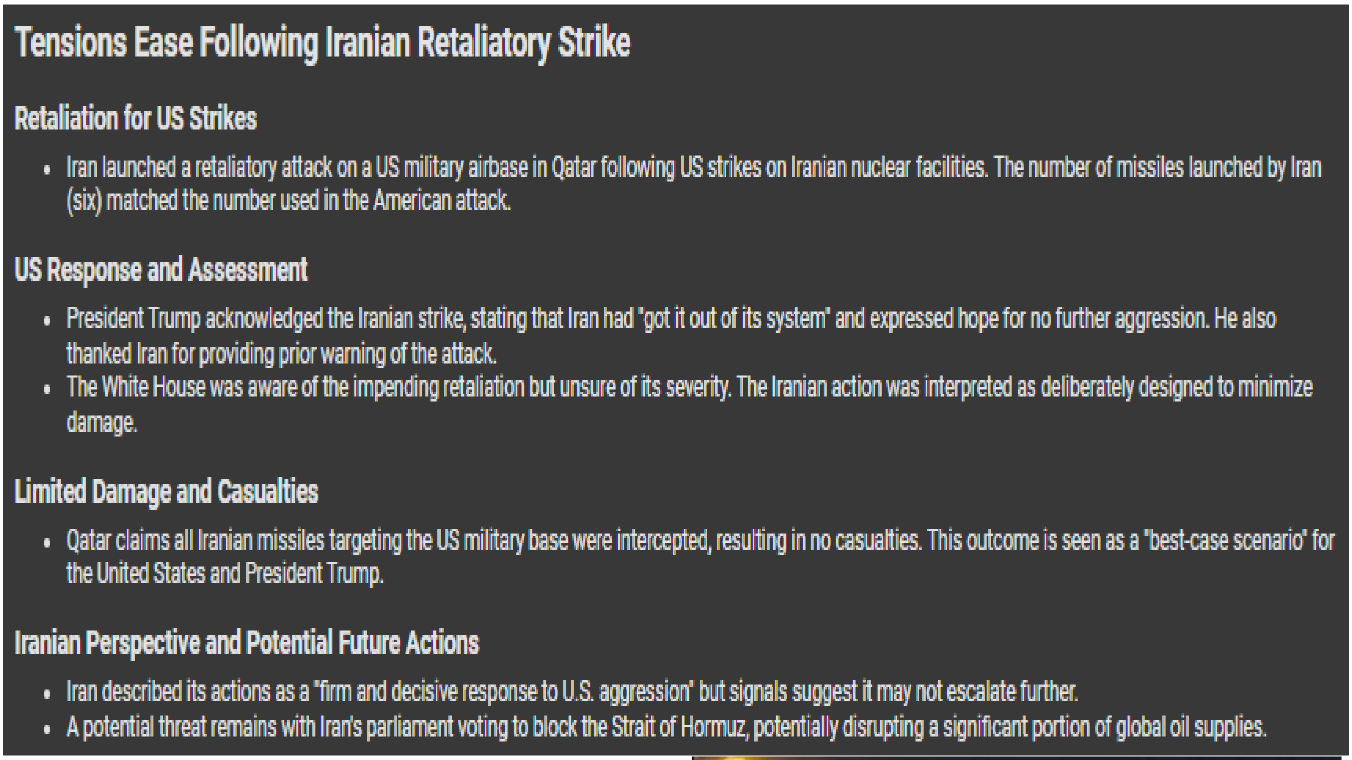
Video Link**:** https://youtu.be/a5QD6oBM8tg

The video explains the escalating conflict between Iran and Israel, highlighting intense moments leading up to the announcement of a ceasefire. It features visuals such as military cargo flights in motion, bombs lighting up the night sky, and secured military zones, effectively conveying the gravity of the situation. The summarization model then extracts these keyframes based on motion cues, object presence, and scene changes to deliver a compact yet informative snapshot of the geopolitical tension. This automated process enables quick understanding of complex real-world events while preserving essential visual and contextual elements.

The above summarized report illustrates the capability of the proposed video summarization model to extract concise and contextually accurate information from complex news video content. In this instance, the model successfully identified and organized the key highlights of a news segment discussing the Iranian retaliatory strike following U.S. military actions. Through automatic speech-to-text transcription, followed by text filtering and semantic summarization, the system generated a structured report with distinct sections—covering retaliation motives, U.S. response, damage assessment, and Iran's future stance. This demonstrates how the model can capture critical details while maintaining factual consistency and narrative flow, thereby transforming long-form video content into an easily understandable written summary. The output also reflects the system's strength in context preservation and thematic segmentation. By recognizing the underlying topics such as "retaliation," "assessment," and "future actions," the model effectively condensed a complex geopolitical event into a readable summary (Fig 6) without omitting vital information.

Fig.7, 8, and 9 were extracted by our ML-based summarization model from a news video depicting escalating Iran–Israel tensions. The selected frames—showing missile launches, military readiness, and civilian panic—capture the progression of the conflict. This highlights the model's capability to condense complex real-world events into impactful visual summaries for fast analysis.

Fig. 6 Summarized Report of Iran-Israel Tensions Extracted from News Video Content



Tensions Ease Following Iranian Retaliatory Strike

Retaliation for US Strikes

- Iran launched a retaliatory attack on a US military airbase in Qatar following US strikes on Iranian nuclear facilities. The number of missiles launched by Iran (six) matched the number used in the American attack.

US Response and Assessment

- President Trump acknowledged the Iranian strike, stating that Iran had "got it out of its system" and expressed hope for no further aggression. He also thanked Iran for providing prior warning of the attack.
- The White House was aware of the impending retaliation but unsure of its severity. The Iranian action was interpreted as deliberately designed to minimize damage.

Limited Damage and Casualties

- Qatar claims all Iranian missiles targeting the US military base were intercepted, resulting in no casualties. This outcome is seen as a "best-case scenario" for the United States and President Trump.

Iranian Perspective and Potential Future Actions

- Iran described its actions as a "firm and decisive response to U.S. aggression" but signals suggest it may not escalate further.
- A potential threat remains with Iran's parliament voting to block the Strait of Hormuz, potentially disrupting a significant portion of global oil supplies.

Output (Extracted images):

Fig .7 Civilian Panic During Missile Launch Alert

Fig. 8 Military Base Readiness Amid Rising Tensions

Fig. 9 Missile Launch Captured from Conflict Video

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C. *Sport Video Summary*

Video Link**:** https://youtu.be/Ar4LrKcwh3w

The summarization (Fig 10) effectively highlights key match moments, including goals, celebrations, and decisive penalty shootouts. The extracted keyframes capture intense gameplay and emotional fan reactions, offering a concise visual representation of the match's dramatic progression. Additionally, the model ensures that redundant or less impactful scenes are filtered out using similarity analysis, preserving only the most significant highlights. This results in an efficient summary that maintains both the emotional depth and chronological flow of the match. Such summarization can be particularly useful for generating automated sports highlights, analytical reviews, and audience engagement.

Generated Summary:

Argentina's World Cup Victory

Key Players and Moments

- **Lionel Messi:** Integral to Argentina's success, scoring crucial goals, including penalties, and leading the team's attack. His performance throughout the tournament cemented his legacy.
- **Kylian Mbappé:** A formidable opponent, scoring multiple goals, including a penalty and a volley, showcasing his skill and determination.
- **Angel di Maria:** Scored a penalty to give Argentina an early lead in a crucial match. His presence on the field impacted the opposition's ability to compete.
- **Alvarez:** Played a challenging role in support of Messi.
- **Edinson Cavani:** Scored a goal in a match, contributing to Argentina's overall performance.
- **Kingsley Coleman:** Scored the opening goal in a match.

Match Highlights

- **Argentina vs. Archonsena:** Argentina secured a 3-victory.
- **Argentina's Victory Over France:** After a 3-3 draw, Argentina won on penalties (4-2) to claim the World Cup title.
- **Penalty Shootouts:** Critical moments throughout the tournament, with Messi converting penalties and contributing significantly to Argentina's success.
- **Argentina vs. Mexico:** Argentina secured a 3-1 victory, with Messi scoring a penalty.

Overall Tournament Summary

- **Argentina's Triumph:** Argentina won their first World Cup since 1990, marking a historic achievement.
- **Scaloni's Influence:** Coach Scaloni played a crucial role in guiding the team to victory.
- **Celebration:** The victory sparked widespread celebrations and cemented the players' legacy as heroes.

Fig. 10 Match Highlights Extracted from Sports Video Content

Output (Extracted images):

Fig. 11 Match Emblem from Opening Ceremony Frame

Fig. 12 Key Goal Moment Captured by Argentina

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Fig. 13 Fans Celebrations During Late Substitution

Fig.11, 12, and 13 represent keyframes automatically extracted from the FIFA World Cup Final video between Argentina and France. Figure 11 highlights the official match emblem from the opening ceremony, marking the significance and global attention surrounding the final. Figure 12 captures a decisive goal moment by Argentina, where player positioning and crowd engagement are clearly visible. Figure 13 shows the electrifying response of Argentinian fans during a late substitution, reflecting the emotional stakes of the match. These frames were selected by the ML-based summarization model using visual context, scoreboard changes, and motion density, ensuring that high-energy, meaningful scenes are retained. The selection offers a compact, impactful summary of the match's flow from formal opening to critical gameplay and fan celebrations.

V. CONCLUSION

This paper introduces a novel hybrid framework for video summarization that effectively combines computer vision techniques with large language models to produce comprehensive and user-friendly summaries. Unlike traditional methods that typically generate only keyframes or plain text, this approach integrates images, text, diagrams, and speech into pedagogically structured notes, enhancing clarity, retention, and usability across various applications, including education, corporate training, and media. For visual summarization, the system leverages OpenCV for frame extraction, YOLO for detecting important objects and actions, and SSIM to eliminate redundant frames, ensuring that only significant and diverse frames are retained as key highlights. For textual summarization, a pre-trained BART model generates coherent and context-aware summaries from video transcripts or audio content, while a Large Language Model (LLM) wrapping mechanism formats the output into clean, structured notes. The integration of these multimodal elements into a unified, structured format represents a significant advancement over conventional summarization techniques, which often provide unstructured outputs or lack cross-modal synthesis. The proposed system's modularity, scalability, and demonstrated capability in generating both high-quality visual highlights and semantically rich, well-formatted textual summaries validate its potential to significantly improve accessibility, comprehension, and learning efficiency Authors and Affiliations

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